

MAYUKH DAS

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EDUCATION

Australian National University, Master of Computing

Feb 2025 – Present

Relevant coursework: Deep Learning (CNNs, RNNs, Transformers, Generative Models), Advanced ML (Diffusion Models, LLMs), Human-Computer Interaction, Software Engineering, Software Construction (Design Patterns, Android), Structured Programming (Java), Logic and Mathematics.

Manipal Institute of Technology, B.Tech, Automobile Engineering

Aug 2017 – Sep 2021

SKILLS

Languages: Python, Java, C#, JavaScript, SQL

ML / AI: Deep Learning, Transformers, Diffusion Models, LLMs, Explainable AI (XAI), Supervised & Unsupervised Learning

Frameworks & Tools: PyTorch, scikit-learn, LLM APIs (OpenAI / GPT), Git, Jupyter, .NET, MVC, React, Node (MERN), HTML, CSS. Rated Advanced in UX Design and Machine Learning at Accenture.

WORK EXPERIENCE

Ecoi Pty Limited

Jul 2026 – Present

Systems Architect · ANU Computing Internship (COMP8830)

- Delivering Phase 1.3 of the **Sovereign AI Community of Practice (SAICOP)**: conceptual architecture, platform assessment and shortlisting, and a Proof of Architecture spanning six workstreams.
- Designed a quality-weighted contribution scoring model resolving two competing stakeholder requirements: incentivising participation while maintaining signal-to-noise in community output.

Australian National University

Jul 2026 – Present

Tutor, COMP3900 Human-Computer Interaction

- Teach weekly tutorials, guiding students through design methods, usability evaluation, and semester project work; assess student work and provide written feedback against course rubrics.

Accenture

Jan 2022 – Feb 2025

Advanced Application Engineering Analyst

- Built **Ingrain**, an AIOps tool generating predictive ML pipelines for automated IT ticket resolution, significantly reducing manual triage effort.
- Integrated Explainable AI (XAI) and dynamic model performance reporting to improve client interpretability of predictions; designed and implemented the supporting user interfaces from scratch.
- Built **Quasar++**, a ChatGPT API-based file-querying chatbot, enabling natural-language access to internal documents.
- Rated Advanced in UX Design and Machine Learning under Accenture's internal competency framework.

Cognizant

Oct 2021 – Jan 2022

Programmer Analyst

- Designed and implemented full-stack web systems using .NET, MVC, React, HTML, CSS, and JavaScript, building all UI from scratch; delivered a Hospital Logistics System to manage COVID-19 operational workflows.

PROJECTS

Transformer's Revenge — Stress-Testing minGRU Against Attention — PyTorch · CUDA · Experimental Design

- Designed a 27-experiment study (3 architectures × 3 tasks × 3 sequence lengths) testing published claims that minGRU matches Transformer performance at linear complexity; derived predictions from architectural analysis before running experiments.
- Implemented the Transformer baseline (RoPE, RMSNorm) and built the comparison framework enforcing an identical training protocol and shared task specification interface across independent model implementations.

GPT Implementation (nanoGPT) — PyTorch · CUDA · Transformer Architecture

- Implemented a GPT language model from scratch in PyTorch, covering attention mechanisms, positional encoding, and autoregressive text generation.

Diffusion Model Implementation — PyTorch · CUDA · Generative AI

- Built a diffusion model from scratch covering score-based generative modelling, noise scheduling, and denoising mathematics, as part of ANU Advanced ML coursework.

FoodLens — Smart Food Scanner PWA — HTML · CSS · JavaScript · Web APIs · PWA

- Built a Progressive Web App for sustainable food shopping: real-time barcode scanning via BarcodeDetector API, live product lookup via Open Food Facts, allergen/expiry/carbon flagging, and A/B interaction logging for HCI research (UN SDG 12 & 3).

Yuma — Android Social Platform — Java · Android · DAO · Singleton · Factory · Iterator

- Co-built a native Android social media app (post feed, reactions, DMs, admin moderation) applying Singleton, Factory, and Iterator design patterns across a layered DAO architecture with full CSV persistence.

STUDENT ROLES

ANU College of Systems and Society

2025 – Present

Student Ambassador

- Represent the ANU College at recruitment events and school visits; support prospective-student transition and onboarding.

ANU Students' Association (ANUSA)

2025 – Present

Postgraduate Representative, College of Systems and Society

- Elected representative for the CSS postgraduate cohort; advocate for student interests and liaise with faculty and administration.

RESEARCH PUBLICATIONS

- Das, M., Castillo Perez, T., Shetty, D., Hiremath, P., Naik, N., Bhat, R. (2024). *An Overview on the Role of Artificial Intelligence in Modern Advancements of Material Science*. ES General, Vol. 5. DOI 10.30919/esg1183. First author.
- Karthik, A., Das, M. et al. (2021). *Electric Propulsion for Fixed Wing Aircrafts: Classifications, Designs, and Challenges*. Engineered Science, Vol. 16, pp. 129–145. DOI 10.30919/es8d573. Scopus-indexed, SJR Q1.
- Momaya, V.J., Hameed, B.Z., Adhikari, K., Das, M. et al. (2023). *Revolutionizing Organ Donation With Blockchain Technology: Prospects and Challenges in Healthcare*. J. of Computers, Mechanical and Management, Vol. 2(5), pp. 31–38. DOI 10.57159/gadl.jcmm.2.5.23071. Scopus-indexed.
- **Conference presentation:** Work-in-progress form of the Material Science paper, National Conference on Recent Trends in Computers, Mechanical and Management (RTCMM), 2023.